

Hemal Arora

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EDUCATION

Stanford University

B.S. Electrical Engineering (Hardware & Software Track) | GPA: 3.85/4.0

M.S. Electrical Engineering (Planned Coterminial Program)

Stanford, CA

Expected Jun 2027

Expected Jun 2028

- **Coursework:** Robot Dynamics, Manipulation & Control (CS223A/CS225A/AA174A); Robot Perception & Computer Vision (CS227A/CS231N); Reinforcement Learning (CS234/CS238); Machine Learning & Statistical Inference (CS229/EE278); Computer Systems, Digital Design & VLSI (CS107E/EE180/EE271)

EXPERIENCE

Stanford Interactive Perception and Robot Learning Lab (IPRL)

Stanford, CA

Undergraduate Researcher: Advised by Prof. Jeannette Bohg

Mar 2026 – Present

- Co-developed ClothAtlas (submitted to ICRA 2027), a graph-conditioned flow-matching model for 3D deformable state estimation that reconstructs complete garment geometry from partial point clouds, preserves correspondences under occlusion, and generalizes to unseen mesh topologies without retraining.
- Transferred ClothAtlas directly from simulation to real-world garment manipulation without fine-tuning, localizing 85.1% of occluded grasp targets within 10 cm vs. 36.2% for dense descriptor matching.
- Researching failure-aware robot learning methods that retrieve relevant prior interactions from memory and use action-conditioned world models to guide test-time recovery in VLA policies.

Mosaic Intelligence Labs, Inc. (Stealth Surgical Robotics Startup)

San Francisco, CA

Member of Technical Staff: Simulation & Perception

Jun – Sep 2026

- Built physics-based sensor simulators and synthetic data pipelines for an endovascular robotic navigation system, calibrating simulated X-ray and Intravascular Ultrasound imaging with clinical data; trained vision models on ~100K synthetic frames that generalized to real clinical imaging during an animal study.
- Extended GPU-accelerated XPBD Cosserat-rod simulations for realistic guidewire/catheter dynamics, implementing batched vessel-contained rod solving, coaxial instrument coupling, and solver-stability fixes; defined simulation backend requirements for scaling large model-predictive planning rollouts.
- Built a real-time coupled instrument dynamics and imaging simulator, enabling hardware-in-the-loop validation of factor-graph-based localization and state estimation algorithms under partial observations.

Multiply Labs

San Francisco, CA

Robotics Software Engineer

Jun – Dec 2025

- Trained Diffusion Policy and Action Chunking Transformer (ACT) models on ~150 teleoperated demonstrations for precise cartridge insertion into millimeter-scale slots, deployed and evaluated on a UR10e robot.
- Developed a motion generation system for the company's robotic cell therapy manufacturing cluster in NVIDIA Isaac Sim, using cuMotion to execute collision-free trajectories without hard-coded waypoints.
- Developed a production-ready camera SDK with strict mypy typing and 100% test coverage, used for cluster-wide ArUco-based auto-calibration and collecting Unitree G1 demonstration data for VLA fine-tuning.

GE HealthCare & Stanford Magnetic Resonance Systems Research Lab

Stanford, CA

Undergraduate Researcher: Advised by Prof. John Pauly

Jun – Sep 2024

- Designed a 24 GHz mm-wave radar system for non-contact cardiac/respiratory MRI gating, including custom PCBs and SSA/ICA signal processing; first-author abstract accepted to ISMRM 2025 [[Abstract](#)].

SELECTED PROJECTS & AWARDS

- **Autonomous Pickleball-Playing Mobile Manipulator** [[Video](#)]: Built a Franka FR3 + TidyBot++ mobile manipulator that tracks incoming balls, predicts intercepts, repositions its base, and replans velocity-controlled strikes at 100 Hz using damped least-squares IK and velocity-constrained Hermite trajectories.
- **Maximum Likelihood RL for Robotic Manipulation** [[Report](#)]: Extended MaxRL to sparse-reward 7-DOF manipulation, benchmarking against RLOO using goal-conditioned behavior cloning, hindsight experience replay, and policy-gradient fine-tuning.
- **1st Place, OpenAI Prize—TreeHacks 2025** [[Devpost](#)]: Built a real-time EEG-adaptive generative storytelling system with GPT-4o, asynchronously orchestrating narrative, audio, and video generation.

SKILLS

- C++, Python, C, PyTorch, ROS 2, MuJoCo, NVIDIA Isaac Sim, Warp, Newton, OptiX, SystemVerilog, KiCad